

Project Initialization and Planning Phase

Date	14 July 2026
Team ID	XXXXXX
Project Title	Personalized Networking Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal outlines an AI-powered web application that helps professionals generate personalized, context-aware conversation starters for networking events, while offering quick, reliable fact verification and a history of past conversations to support continuous improvement in networking skills.

Project Overview	
Objective	To help professionals generate personalized, context-aware conversation starters and quickly verify facts before and during networking events, improving networking confidence and outcomes.
Scope	Design and development of a Streamlit-based frontend, a FastAPI backend, an NLP pipeline (DistilBERT for theme extraction and GPT-2 for text generation), Wikipedia-based fact verification, and local history/feedback logging for a local single-user deployment.
Problem Statement	
Description	Professionals attending networking events often struggle to identify relevant conversation topics quickly and to verify facts about unfamiliar subjects on the spot, leading to missed opportunities and reduced confidence.
Impact	Solving this helps users start more meaningful conversations, feel better prepared, and build stronger professional connections, while reducing reliance on generic, unreliable last-minute research.
Proposed Solution	
Approach	Use NLP techniques - zero-shot classification with DistilBERT and text generation with GPT-2 - to analyze event/bio inputs and produce tailored conversation starters, paired with a lightweight fact-verification layer built on the Wikipedia API.
Key Features	- Automatic extraction of event themes from free-text event descriptions using DistilBERT zero-shot classification. - Context-aware conversation-starter generation using a GPT-2 text-generation pipeline. - On-demand fact verification of specific topics via the Wikipedia Search & Lookup API. - Local history logging of past conversation starters with thumbs-up/down feedback to support personalization over time.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i3/i5 or higher (no dedicated GPU required)
Memory	RAM specifications	Minimum 4 GB (8 GB recommended)
Storage	Disk space for data, models, and logs	10 GB free disk space
Software		
Frameworks	Python frameworks	FastAPI, Streamlit
Libraries	Additional libraries	Hugging Face Transformers, Wikipedia API wrapper, pytest, httpx
Development Environment	IDE	Visual Studio Code, PyCharm
Data		
Data	Source, size, format	User-entered bio & event description text (real-time input); pre-trained DistilBERT and GPT-2 models used directly, no external training dataset required